DSEH001 · DIGITAL TRANSFORMATION IN HR SYSTEMS

02

UNIT 2 READER

Building and Rolling Out Digital HR

HRIS, electronic HRM, adoption and change

Weeks 4 to 6 | Round 2: The Architecture | Round 3: The Rollout | 12 readings explained

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About this reader

> **How to use it.** This reader is a companion to the set readings, not a replacement for them. For each reading it explains the argument in plain language, lists the ideas you must hold, links them to GreenLeaf and the Disha rounds, flags common confusions, and gives questions to think with. Read the original first, then use this reader to check your understanding. Folder names refer to the course Readings folder.

Unit 2 moves from diagnosis to design and delivery. Its first half is about architecture: what an HR information system is, how its parts should fit together, and how privacy must be built in rather than bolted on. Its second half is about people: why employees adopt or ignore a new system, and how a change program can move them from knowing a system exists to using it and recommending it to others.

The readings come from two research traditions that you should learn to combine. Information systems research (Venkatesh and colleagues, Bondarouk and colleagues) explains adoption through perceptions, conditions and context. Change management (Lewin, Kotter, Prosci ADKAR, Everett Rogers) explains how to lead people through a transition. A strong Round 3 answer uses the first tradition to find where a group is stuck, and the second to choose the fix that matches that stuck point.

READING MAP

Week	Set readings	Folder	Disha link
4	Bondarouk, Parry and Furtmueller (2017); Kavanagh and Johnson (2018) Ch 1 and 3	Unit-2 Core	Round 2
5	Scully-Russ and Torraco (2020) pp. 83 to 84; Prosci ADKAR	Unit-1 Core, Unit-1 Supp.	Round 3
6	Bondarouk et al. (2017) adoption sections; Venkatesh et al. (2003); Manuti and de Palma (2018) Ch 4 and 5	Unit-2 Core	Round 3 and D1 Clinic

The readings, one by one

Week 4 Core

Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *The International Journal of Human Resource Management*, 28(1), 98 to 131.

THE ARGUMENT IN BRIEF

This is a systematic review of forty years of research on electronic HRM. The authors organize what is known about adoption into three groups of factors: technology (how useful and usable the system is), organization (support, resources, culture, the role of the HR function) and people (attitudes, skills and experience of users). They then review the consequences of electronic HRM and find that most studies report operational gains such as speed and lower cost, while evidence on relational and strategic outcomes is weaker and mixed. Their critique of the field is that much research describes rather than explains, and that adoption and consequences are rarely studied together.

IDEAS TO HOLD

- Adoption depends on technology, organization and people factors together.
- Operational outcomes are well documented; relational and strategic outcomes much less so.
- Adoption and consequences should be studied as one chain, not separately.
- Context, including national and organizational culture, shapes both.

CONNECT TO GREENLEAF

Use the three factor grouping to structure your Round 3 diagnosis of who is being left behind. A single factor explanation (for example, 'they need training') is usually incomplete.

EASY TO CONFUSE

Do not cite Bondarouk, Ruël and Parry, *Electronic HRM in the Smart Era*, in place of this article. It is a different work and is background only.

QUESTIONS TO THINK WITH

- For one GreenLeaf segment with low adoption, list one technology, one organization and one people factor that could explain it, with the tab you would check.

Week 4 Core

Kavanagh, M. J., & Johnson, R. D. (2018). Chapters 1 and 3.

THE ARGUMENT IN BRIEF

Chapter 1 traces how HRM and HR information systems evolved together, from record keeping to strategic support, and shows why HR data became a business asset. The Chapter 3 material sets out how a system is planned and built: a structured life cycle that begins with analysis of needs, moves through design and implementation, and continues with maintenance and review. The message for practitioners is that most system failures begin in poor needs analysis, when the organization buys features before it has defined the decisions and users the system must serve.

IDEAS TO HOLD

- HRIS grew from administrative record keeping toward strategic decision support.
- Systems are built through a life cycle: needs analysis, design, implementation, review.
- Needs analysis defines users, decisions and data before vendors are compared.
- Skipping analysis is the most common root of failed implementations.

CONNECT TO GREENLEAF

Your Round 2 blueprint is a needs analysis in miniature. Start from GreenLeaf's users (including Smartphone only and Feature phone staff) and decisions, then choose systems.

QUESTIONS TO THINK WITH

- Write three needs statements for GreenLeaf's frontline that any target architecture must meet.

Week 4 Round 2

Course framing: the three layer architecture (System of Record, Engagement, Insight).

THE ARGUMENT IN BRIEF

Round 2 uses the course's three layer architecture framing. The System of Record is the single authoritative source of employee data (who works here, in what role, on what pay). The Engagement layer is how people interact with HR: self service apps, chatbots, learning platforms. The Insight layer turns data into analysis and decisions. The design rule is one record only: every engagement and insight tool should read from and write to the same system of record, so that the organization never holds two conflicting versions of an employee.

IDEAS TO HOLD

- System of Record: one authoritative employee record.
- Engagement: the channels employees and managers actually touch.
- Insight: analytics and reporting built on the record.
- Two systems of record create duplicate, conflicting data and weaken every layer above.

CONNECT TO GREENLEAF

TechStack_Inventory lists two systems in the Core HRIS category, one on premise and one cloud. Decide which becomes the record and what happens to the other.

EASY TO CONFUSE

This framing is the course's own; credit it to the course, not to Kavanagh. It is also not a maturity ladder: the layers coexist, they are not stages.

QUESTIONS TO THINK WITH

- Which GreenLeaf system would you choose as the System of Record, and which evidence (integration, deployment, adoption) supports it?

Week 4 Round 2

Cavoukian, A. (2011). *Privacy by Design: The 7 foundational principles*. Information and Privacy Commissioner of Ontario. ISO/IEC 27701:2025.

THE ARGUMENT IN BRIEF

Privacy by Design holds that privacy should be built into systems from the start and set as the default, rather than added after a problem appears. Its principles include being proactive rather than reactive, making privacy the default setting, embedding it into the design, protecting data through its whole life cycle, and keeping processing visible and user centered. ISO/IEC 27701 sets out a privacy information management system that gives organizations an auditable structure for these practices. Its 2019 edition was an extension of ISO/IEC 27001; the revised edition published in October 2025 is a standalone standard, so certification no longer requires ISO/IEC 27001 first. In the Indian setting, these ideas are the practical route to meeting DPDP duties such as purpose limitation and data minimization.

IDEAS TO HOLD

- Proactive, not reactive; privacy as the default setting.
- Consent, purpose limitation and data minimization designed into each flow.
- Protection across the full life of the data, including deletion.
- ISO/IEC 27701 (2025 edition, now standalone) provides an auditable management system for privacy.

CONNECT TO GREENLEAF

For each layer of your Round 2 architecture, state what data it collects, for what purpose, where it is stored and when it is deleted. Data residency in TechStack_Inventory is a design input, not an afterthought.

EASY TO CONFUSE

Compliance first (Round 1) audits what already exists. Privacy by Design (Round 2) shapes what you are about to build.

QUESTIONS TO THINK WITH

- Name one GreenLeaf data flow that would fail data minimization as it stands today.

Week 4 Round 2

India Stack: Aadhaar eKYC, UPI and DigiLocker as public digital infrastructure.

THE ARGUMENT IN BRIEF

India Stack is a set of public digital rails: identity verification through Aadhaar eKYC, payments through UPI, and verified document storage through DigiLocker. For HR, these rails can remove paper from onboarding and payments for a frontline workforce that may lack laptops or formal banking habits. Their attraction is reach and low cost; their risk is that identity and financial data are highly sensitive, so every use must meet the DPDP Act's tests of purpose, consent and security.

IDEAS TO HOLD

- Public rails can make onboarding and payment possible on a basic phone.
- Reach and cost favor them for large frontline populations.
- Sensitive identity and payment data raise the bar for consent and security.

> **CONNECT TO GREENLEAF**
>
> Compare an India Stack approach with a mobile first HRMS for GreenLeaf's Tier 3 staff. Which one reaches Feature phone users at all?

> **QUESTIONS TO THINK WITH**
>
> - What would an onboarding flow using DigiLocker remove, and what new risk would it add?

Week 5 Core

Scully-Russ, E., & Torraco, R. (2020). Pages 83 to 84.

THE ARGUMENT IN BRIEF

The Week 5 pages turn from the drivers of change to what they mean for learning and development. As tasks change, workers need continuous learning rather than one time training, and organizations need ways to help people build new capabilities while they work. The authors connect this to a wider shift in the employment relationship: responsibility for development is increasingly shared between employer and worker, which can leave less protected workers behind if support is not designed for them.

IDEAS TO HOLD

- Changing tasks require continuous, work embedded learning.

- Responsibility for development is shifting toward workers.
- Less protected workers risk being left behind without designed support.

CONNECT TO GREENLEAF

Read LMS_Learning by Mode and by employee segment. Ask whether GreenLeaf's learning offer reaches the people whose roles are changing fastest.

QUESTIONS TO THINK WITH

- Is self paced mobile learning a fair default for a Tier 3 workforce? What evidence would settle the question?

Week 5 **Round 3**

Hiatt, J. M. (2006). ADKAR: A *model for change in business, government and our community*. Prosci.

THE ARGUMENT IN BRIEF

ADKAR describes individual change as five elements in a fixed order: Awareness of the need for change, Desire to take part, Knowledge of how to change, Ability to put that knowledge into practice, and Reinforcement to sustain it. Its practical value is diagnostic. Find the first element where a person or group is stuck, and fix that element; effort spent further along the sequence is wasted. Knowledge (knowing how) and Ability (being able to do it in real conditions) are often confused, and the gap between them is where many training programs fail.

IDEAS TO HOLD

- Five elements in order: Awareness, Desire, Knowledge, Ability, Reinforcement.
- Diagnose the first stuck element, then match the fix to it.
- Knowledge is knowing how; Ability is doing it in real conditions with real constraints.
- Reinforcement keeps people from slipping back after the launch energy fades.

CONNECT TO GREENLEAF

Map Adoption_Telemetry's Adoption_Stage to ADKAR with care. A trained employee with low ESS use may be stuck at Ability or Desire, not Knowledge.

EASY TO CONFUSE

An ADKAR Ability barrier is not the same as a digital inclusion Access barrier. If a person has no suitable device, no amount of practice builds Ability.

QUESTIONS TO THINK WITH

- For a Smartphone only, low literacy employee in Tier 3 who completed training but rarely logs in, which ADKAR element is the likeliest block? What would you check first?

Week 5 **Round 3**

Lewin, K. (1947). Frontiers in group dynamics. *Human Relations*, 1(1), 5 to 41. Kotter, J. P. (1996). *Leading Change*. Harvard Business School Press.

THE ARGUMENT IN BRIEF

Lewin's model treats change as three phases: unfreeze the current state, move to the new one, and refreeze so that the new way becomes stable. A pilot is a controlled unfreeze: it loosens existing habits in a small, protected setting where learning is cheap. Kotter's eight steps describe the leadership work of large scale change. Round 3 draws on Steps 3 to 5: develop a vision and strategy, communicate it widely and repeatedly, and empower people to act by removing obstacles. Kotter sees these steps as needing speed once urgency (Step 1) and a guiding coalition (Step 2) are in place.

IDEAS TO HOLD

- Lewin: unfreeze, change, refreeze; a pilot unfreezes safely.
- Refreezing is what turns a successful pilot into normal practice.
- Kotter Steps 3 to 5: vision, communication, empowerment by removing obstacles.
- The two models answer different questions: Lewin describes phases, Kotter prescribes leadership actions.

CONNECT TO GREENLEAF

If you recommend a pilot in Round 3, say what it will unfreeze, how you will judge it, and what would trigger refreezing at scale.

EASY TO CONFUSE

Lewin's unfreeze is not the same as Kotter's urgency step, although both prepare people for change.

QUESTIONS TO THINK WITH

- Which Kotter step does a Tier 3 champion network serve? Which obstacle does it remove?

Week 6 Core

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425 to 478.

THE ARGUMENT IN BRIEF

Venkatesh and colleagues compared eight earlier models of technology acceptance and combined them into the Unified Theory of Acceptance and Use of Technology (UTAUT). Four factors explain intention and use: performance expectancy (the belief that the system helps me do my job), effort expectancy (how easy it is), social influence (whether important others think I should use it) and facilitating conditions (whether the infrastructure and support exist). The effects are moderated by gender, age, experience and whether use is voluntary. Facilitating conditions act directly on use, not only through intention.

IDEAS TO HOLD

- Four determinants: performance expectancy, effort expectancy, social influence, facilitating conditions.
- Moderators: gender, age, experience and voluntariness of use.
- Social influence matters more when use is mandatory.
- Facilitating conditions shape actual use directly, which links UTAUT to device and connectivity access.

CONNECT TO GREENLEAF

UTAUT's facilitating conditions connect to the Access layer of digital inclusion. In Adoption_Telemetry, read usage by Device_Access before you blame motivation.

QUESTIONS TO THINK WITH

- Which UTAUT factor would a peer champion program act on? Which one would a device subsidy act on?

Week 6 Round 3

Rogers, E. M. (2003). *Diffusion of Innovations* (5th ed.). Free Press.

THE ARGUMENT IN BRIEF

Everett Rogers explains how innovations spread through a social system over time. Adopters fall into categories: innovators, early adopters, early majority, late majority and laggards. Early adopters are especially important because they are respected opinion leaders whom others watch. Five attributes of an innovation affect how fast it spreads: relative advantage, compatibility with existing values and practice, complexity, trialability and observability. Diffusion works through networks: people adopt when trusted peers show that the innovation works.

IDEAS TO HOLD

- Adopter categories from innovators to laggards.
- Early adopters are opinion leaders; recruit them as champions.
- Five attributes speed or slow spread, including trialability and observability.
- Spread runs through social networks, not only official channels.

CONNECT TO GREENLEAF

A Rogers move recruits respected peers in each store or warehouse to model use. It works where the barrier is social, not where the barrier is a missing device.

EASY TO CONFUSE

Do not confuse Everett M. Rogers with David L. Rogers, author of *The Digital Transformation Playbook* (Unit-1 Core), who writes about business strategy.

QUESTIONS TO THINK WITH

- Which of Rogers' five attributes does a free trial week change? Which does a public leaderboard change?

Week 6 Core

Manuti, A., & de Palma, P. D. (2018). Chapters 4 and 5.

THE ARGUMENT IN BRIEF

The Week 6 chapters ask how HR practices themselves change when cognitive technologies arrive, and how digital tools can become allies of HR rather than replacements for it. Using case evidence, the authors argue that tools for recruiting, learning and communication succeed when HR frames them within the organization's culture and its people strategy. Where tools are introduced as standalone technical projects, adoption is shallow and the employment relationship can suffer.

IDEAS TO HOLD

- HR practices are redesigned, not only automated, by cognitive technology.
- Digital tools work best as allies embedded in culture and people strategy.
- Standalone technical rollouts tend to produce shallow adoption.

> **CONNECT TO GREENLEAF**
>
> Use this to test your Round 3 options. Does each option change only the tool, or also the practice and the culture around it?

> **QUESTIONS TO THINK WITH**
>
> - Pick one Round 3 option. What part of HR practice, not technology, would have to change for it to succeed?

Week 6 Round 3

Nudge and gamification in workplace technology.

THE ARGUMENT IN BRIEF

Nudges (reminders, defaults, simple prompts) and gamification (points, badges, leaderboards) can raise activity quickly. Their limitation is well documented: extrinsic rewards create activity without necessarily creating capability or lasting commitment, and activity can fall when rewards stop. They also risk embarrassing people who struggle, for example on a public leaderboard. Used well, they reinforce behavior that people are already able and willing to perform.

IDEAS TO HOLD

- Nudges and rewards change activity faster than they change capability.

- Activity can collapse when the reward is removed.
- Public rankings can shame those facing Access or Skills barriers.
- Best used as Reinforcement after Ability is in place.

CONNECT TO GREENLEAF

A rise in ESS logins after a points scheme is not evidence of adoption. Compare logins with Self_Service_Txns_per_Month and Adoption_Stage.

EASY TO CONFUSE

Gamification is not diffusion. Rogers spreads use through trusted people; gamification rewards individual activity.

QUESTIONS TO THINK WITH

- What evidence would show that a gamified campaign built capability rather than clicks?

Round 3: matching the fix to the stuck point

If the barrier is…	The framework that fits	A fix that matches
No suitable device or connection	Digital inclusion: Access	Device access, shared kiosks, feature phone channels
Cannot use the system in real conditions	ADKAR: Ability	Coached practice on the job, simplified flows
Does not see why to change	ADKAR: Awareness or Desire	Evidence, manager messages, peer stories
Nobody nearby uses it	Rogers diffusion	Respected local champions
Untested at scale	Lewin: controlled unfreeze	A bounded pilot with clear success criteria
Early use fading	ADKAR: Reinforcement; nudges	Recognition, defaults, follow up

Key terms

Electronic HRM	The use of web and mobile technology to deliver HR practices and policies.
Facilitating conditions	In UTAUT, the infrastructure and support that make use possible.
Guiding coalition	Kotter's Step 2: a group with enough power and credibility to lead change.
Purpose limitation	Using personal data only for the purpose stated when it was collected.
System of Record	The single authoritative source of employee data.
Voluntariness	Whether system use is optional; it changes the weight of social influence.

References

Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *The International Journal of Human Resource Management*, 28(1), 98 to 131.

Cavoukian, A. (2011). *Privacy by Design: The 7 foundational principles*. Information and Privacy Commissioner of Ontario.

Hiatt, J. M. (2006). ADKAR: A *model for change in business, government and our community*. Prosci.

ISO/IEC 27701:2025. *Information security, cybersecurity and privacy protection: Privacy information management systems*. ISO. (Replaces the 2019 edition.)

Kavanagh, M. J., & Johnson, R. D. (Eds.). (2018). *Human resource information systems* (4th ed.). Sage.

Kotter, J. P. (1996). *Leading change*. Harvard Business School Press.

Lewin, K. (1947). Frontiers in group dynamics. *Human Relations*, *1*(1), 5 to 41.

Manuti, A., & de Palma, P. D. (2018). *Digital HR*. Palgrave Macmillan.

Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.

Scully-Russ, E., & Torraco, R. (2020). The changing nature and organization of work. *Human Resource Development Review*, *19*(1), 66 to 93.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, *27*(3), 425 to 478.

The three layer architecture is the course's own framing and is not credited to Kavanagh. Lewin, Kotter, Rogers and Cavoukian are named in the round briefs; their original works are listed so you can trace each idea to its source.